

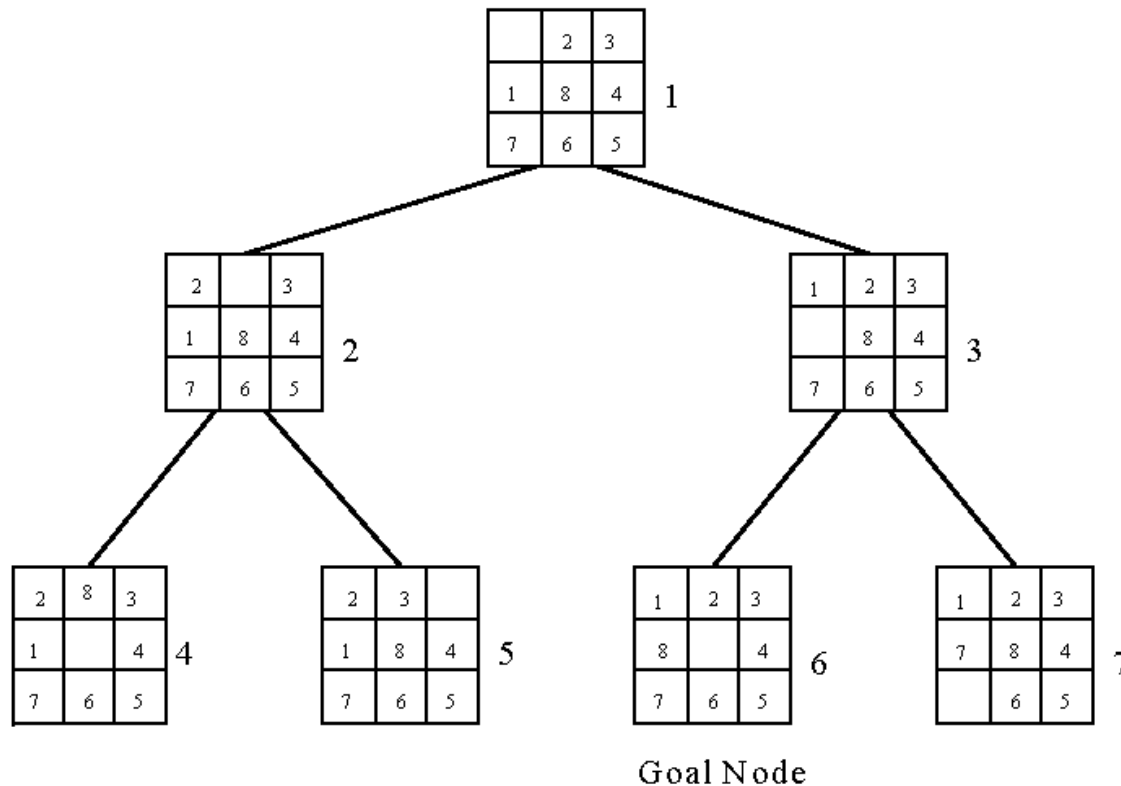
人工智慧導論



The Searching Strategies

Breadth-First Search

8-puzzle problem



- The breadth-first search uses a **queue** to hold all expanded nodes.

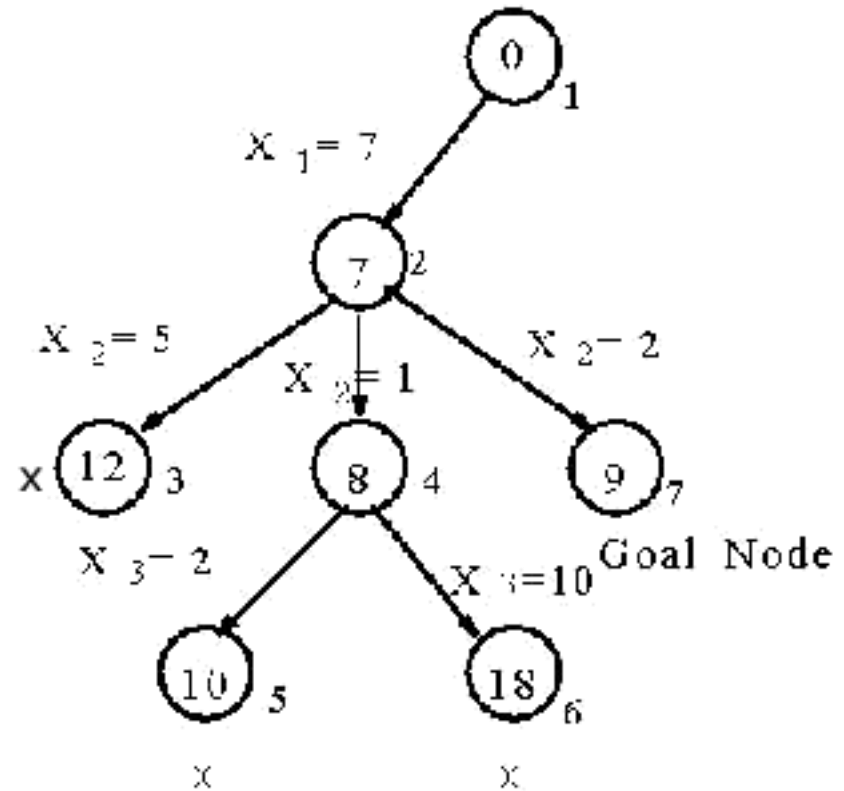
Depth-First Search

- e.g. sum of subset problem

$$S = \{7, 5, 1, 2, 10\}$$

$$\exists S' \subseteq S \ni \text{sum of } S' = 9 ?$$

- A **stack** can be used to guide the depth-first search.



A sum of subset problem
solved by depth-first search.³

Hill Climbing

- A variant of **depth-first search**

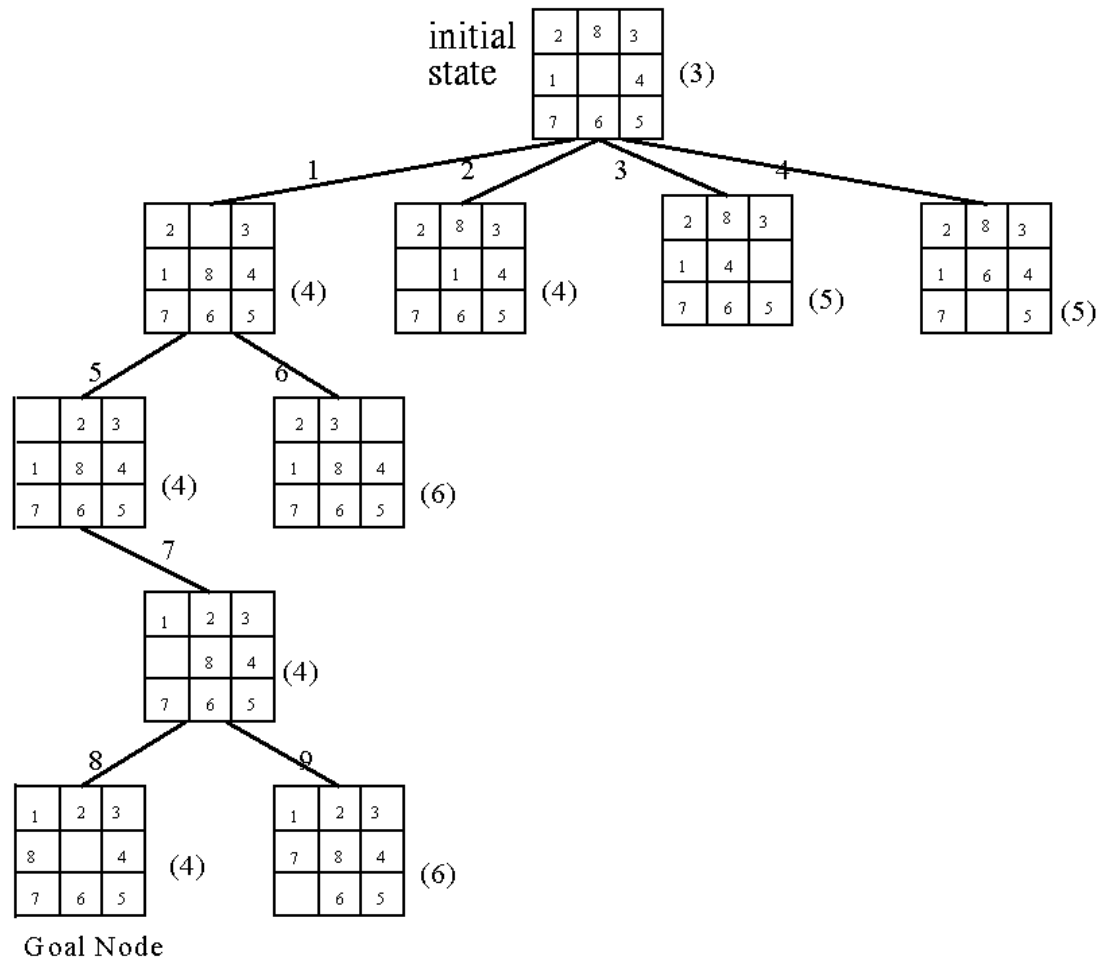
The method selects the locally optimal node to expand.

- e.g. **8-puzzle problem**

evaluation function $f(n) = d(n) + w(n)$

where $d(n)$ is the depth of node n

$w(n)$ is # of misplaced tiles in node n .



An 8-puzzle problem solved by a hill climbing method.

Best-First Search

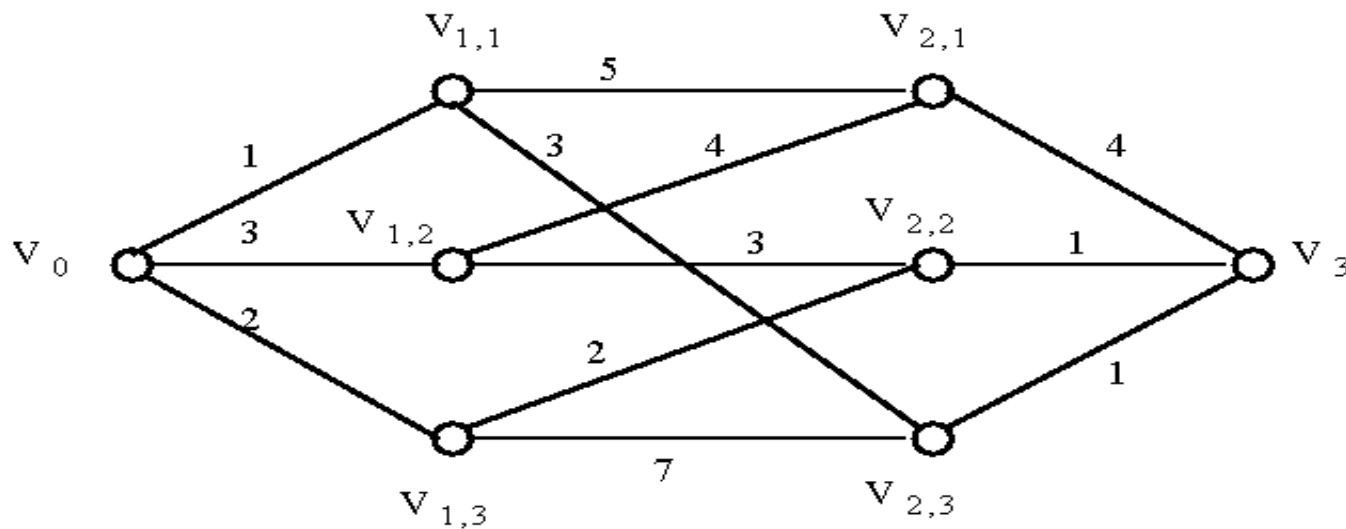
- ❑ Combining depth-first search and breadth-first search.
- ❑ Selecting the node with the best estimated cost among all nodes.
- ❑ This method has a global view.

Best-First Search Scheme

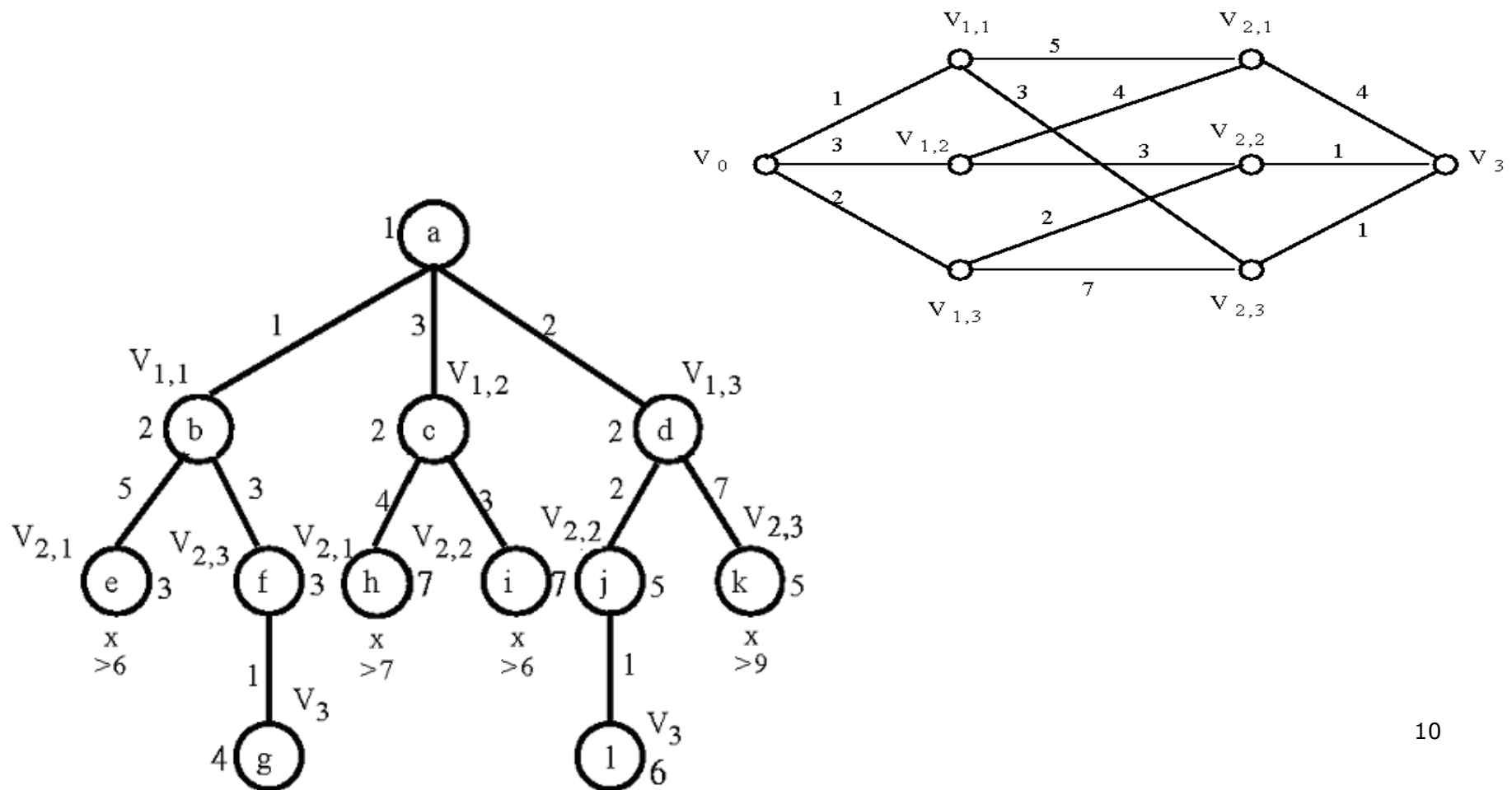
- Step1:** Form a one-element list consisting of the root node.
- Step2:** Remove the first element from the list. Expand the first element. If one of the descendants of the first element is a goal node, then stop; otherwise, add the descendants into the list.
- Step3:** Sort the entire list by the values of some estimation function.
- Step4:** If the list is empty, then failure. Otherwise, go to Step 2.

Branch-and-Bound Strategy

- This strategy can be used to efficiently solve optimization problems.
- e.g. A multi-stage graph searching problem.



▣ Solved by branch-and-bound



Traveling Salesman Problem

- It is NP-complete.
- A cost matrix

j i	1	2	3	4	5	6	7
1	∞	3	93	13	33	9	57
2	4	∞	77	42	21	16	34
3	45	17	∞	36	16	28	25
4	39	90	80	∞	56	7	91
5	28	46	88	33	∞	25	57
6	3	88	18	46	92	∞	7
7	44	26	33	27	84	39	∞

-
- A **reduced cost matrix** can be obtained:
subtract a constant from each row and each column respectively such that each row and each column contains at least one zero.
 - **Total cost subtracted** is a lower bound of our solution.

□ A reduced cost matrix

i \ j	1	2	3	4	5	6	7	
1	∞	0	90	10	30	6	54	(-3)
2	0	∞	73	38	17	12	30	(-4)
3	29	1	∞	20	0	12	9	(-16)
4	32	83	73	∞	49	0	84	(-7)
5	3	21	63	8	∞	0	32	(-25)
6	0	85	15	43	89	∞	4	(-3)
7	18	0	7	1	58	13	∞	(-26)

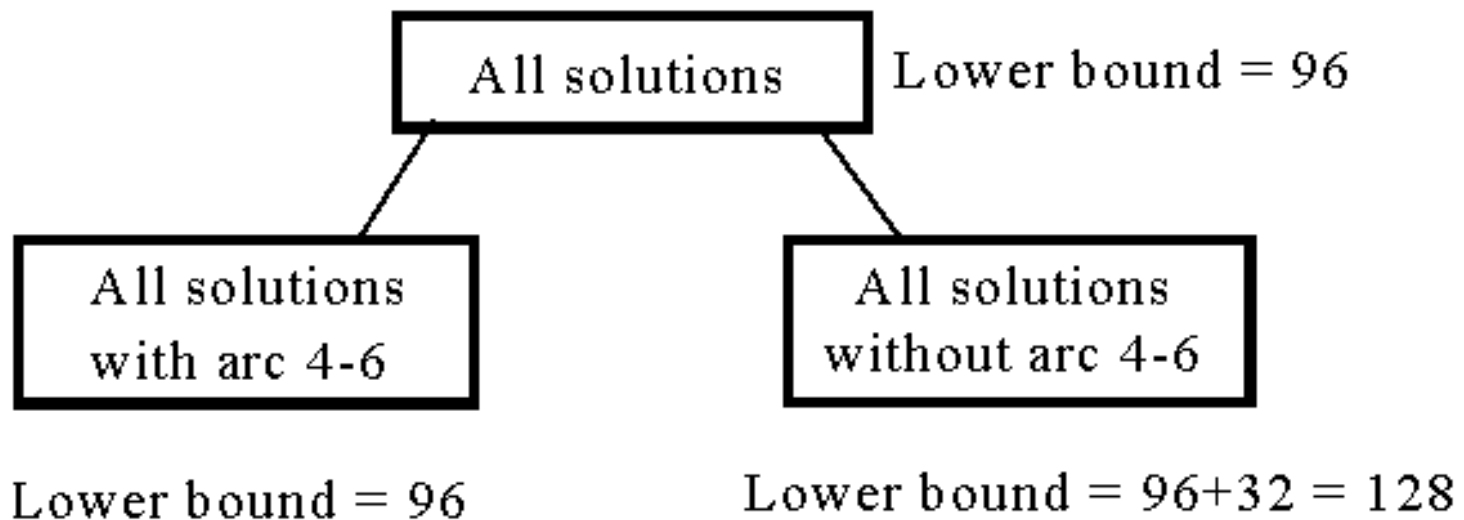
Reduced: **84**

□ Another reduced matrix

j	1	2	3	4	5	6	7
i							
1	∞	0	83	9	30	6	50
2	0	∞	66	37	17	12	26
3	29	1	∞	19	0	12	5
4	32	83	66	∞	49	0	80
5	3	21	56	7	∞	0	28
6	0	85	8	42	89	∞	0
7	18	0	0	0	58	13	∞
			(-7)	(-1)			(-4)

Total cost reduced: $84+7+1+4 = 96$ (lower bound)

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- The highest level of a decision tree:

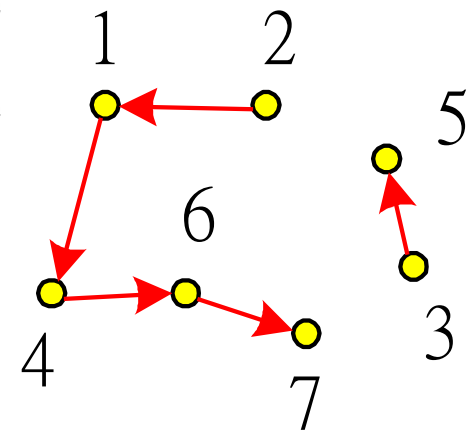
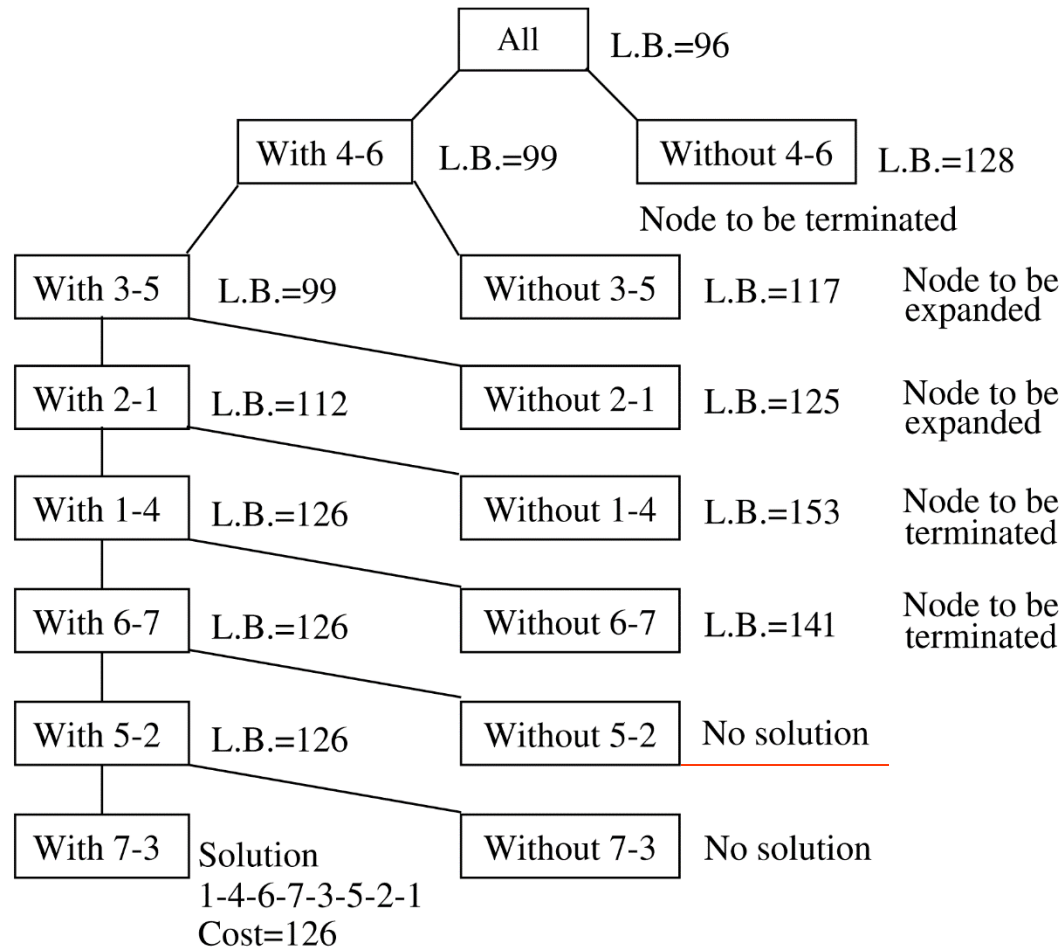


- If a tour does not include arc 4-6, then it must include some other arc emanates from 4. The arc with the least cost emanates from 4 is arc 4-1 whose cost is 32.

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- A reduced cost matrix if arc (4,6) is included in the solution.

j i	1	2	3	4	5	7
1	∞	0	83	9	30	50
2	0	∞	66	37	17	26
3	29	1	∞	19	0	5
5	3	21	56	7	∞	28
6	0	85	8	∞	89	0
7	18	0	0	0	58	∞

Arc (6,4) is changed to be infinity since it can not be included in the solution.



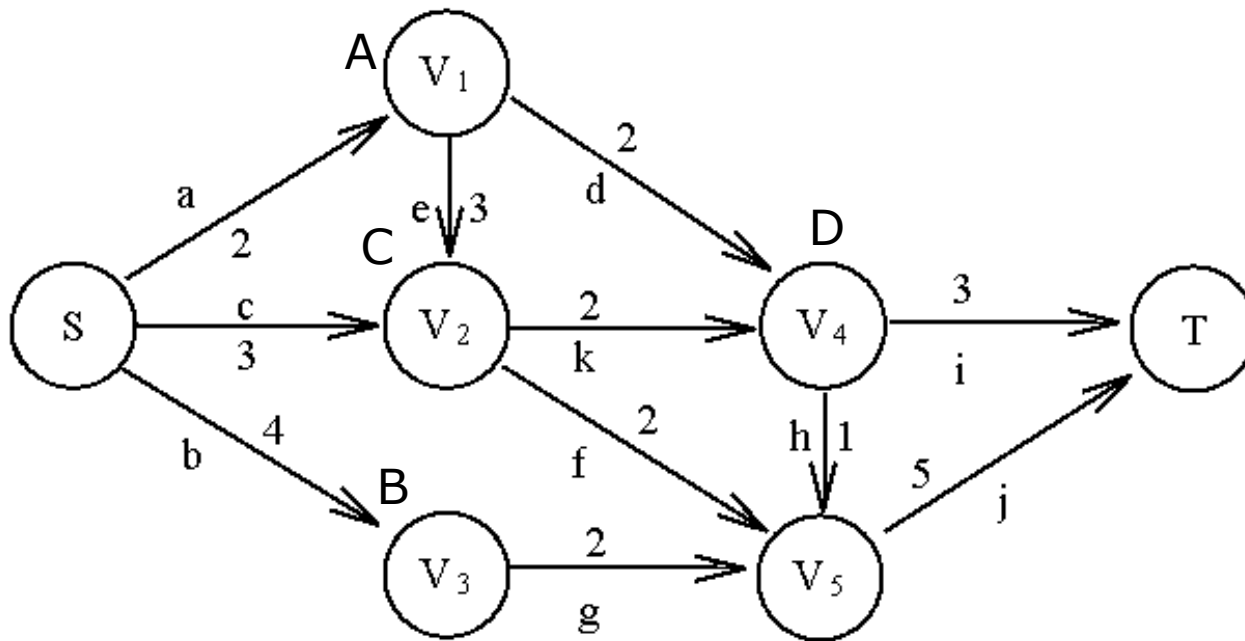
A branch-and-bound solution of a traveling salesperson problem.

The A* Algorithm

- Used to solve optimization problems.
- Using the best-first strategy.
- If a feasible solution (goal node) is obtained, then it is optimal and we can stop.
- Cost function of node n : $f(n)$
 $f(n) = g(n) + h(n)$
 $g(n)$: cost from root to node n .
 $h(n)$: estimated cost from node n to a goal node.
 $h^*(n)$: “real” cost from node n to a goal node.
- If we guarantee $h(n) \leq h^*(n)$, then
 $f(n) = g(n) + h(n) \leq g(n) + h^*(n) = f^*(n)$

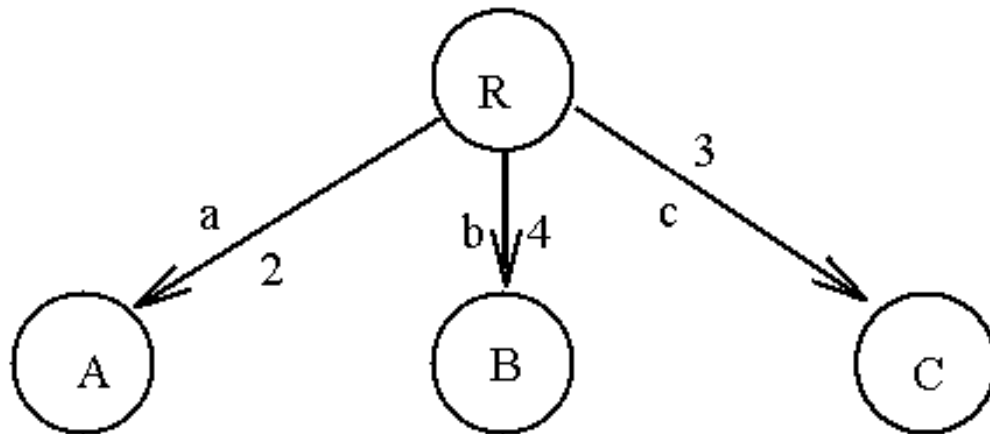
An Example for A* Algorithm

- A Graph to Illustrate A* Algorithm.



- Stop iff the selected node is also a goal node.

□ Step 1:



$$g(A)=2$$

$$g(B)=4$$

$$g(C)=3$$

$$h(A)=\min\{2,3\}=2$$

$$h(B)=\min\{2\}=2$$

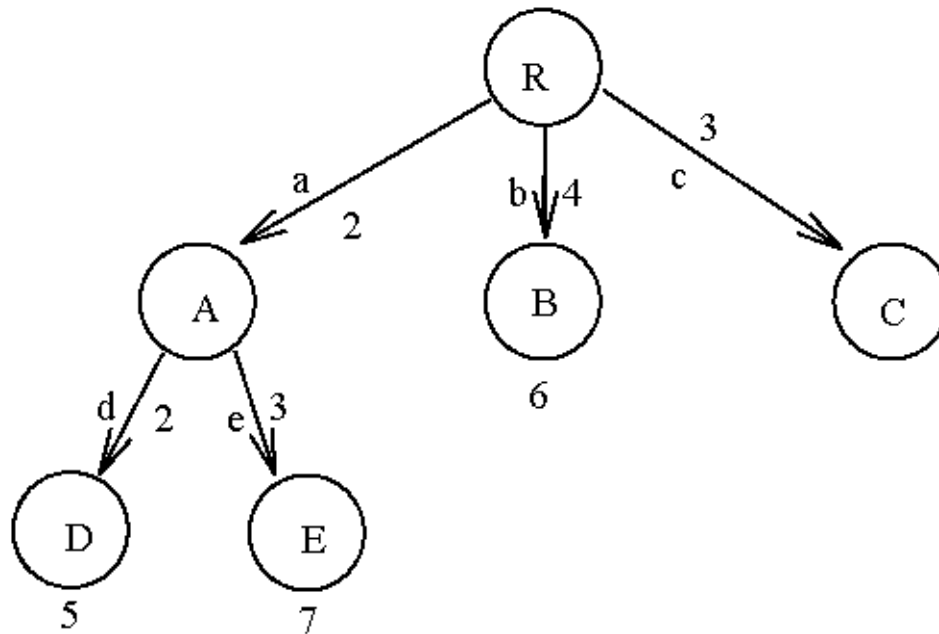
$$h(C)=\min\{2,2\}=2$$

$$f(A)=2+2=4$$

$$f(B)=4+2=6$$

$$f(C)=3+2=5$$

■ Step 2: Expand node A.



$$g(D)=2+2=4$$

$$h(D)=\min\{3,1\}=1$$

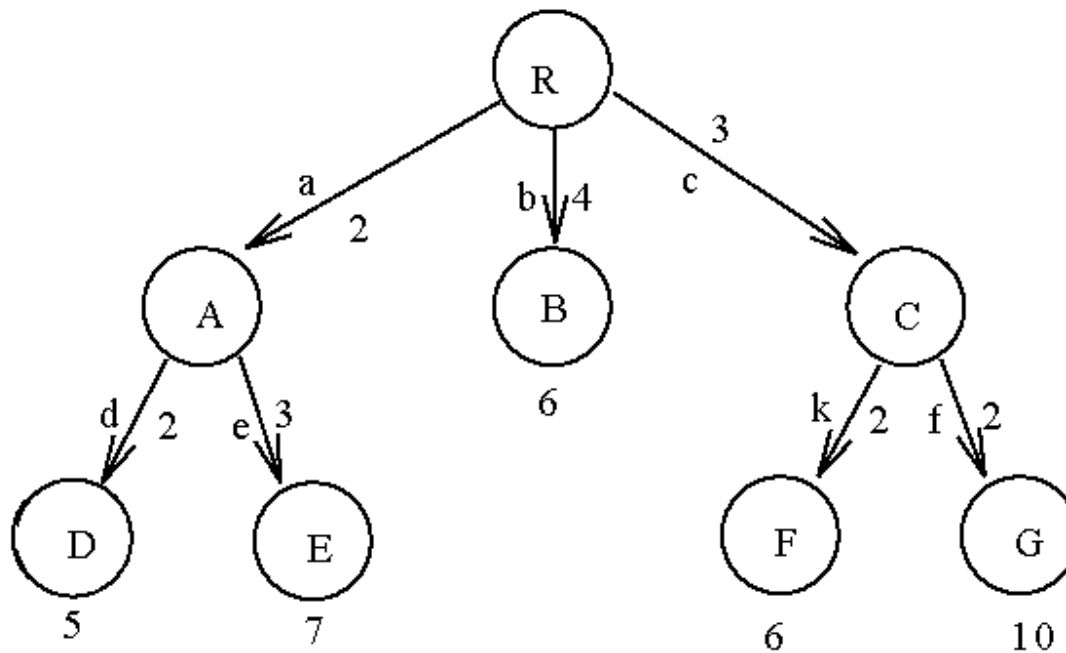
$$f(D)=4+1=5$$

$$g(E)=2+3=5$$

$$h(E)=\min\{2,2\}=2$$

$$f(E)=5+2=7$$

□ Step 3: Expand node C.



$$g(F)=3+2=5$$

$$h(F)=\min\{3,1\}=1$$

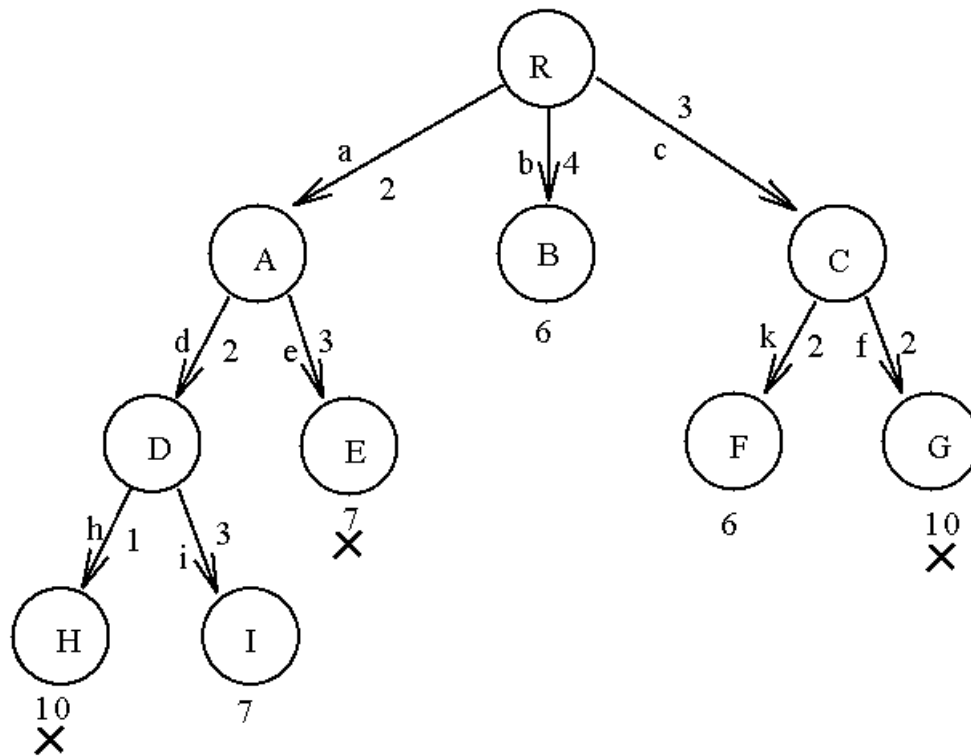
$$f(F)=5+1=6$$

$$g(G)=3+2=5$$

$$h(G)=\min\{5\}=5$$

$$f(G)=5+5=10$$

■ Step 4: Expand node D.

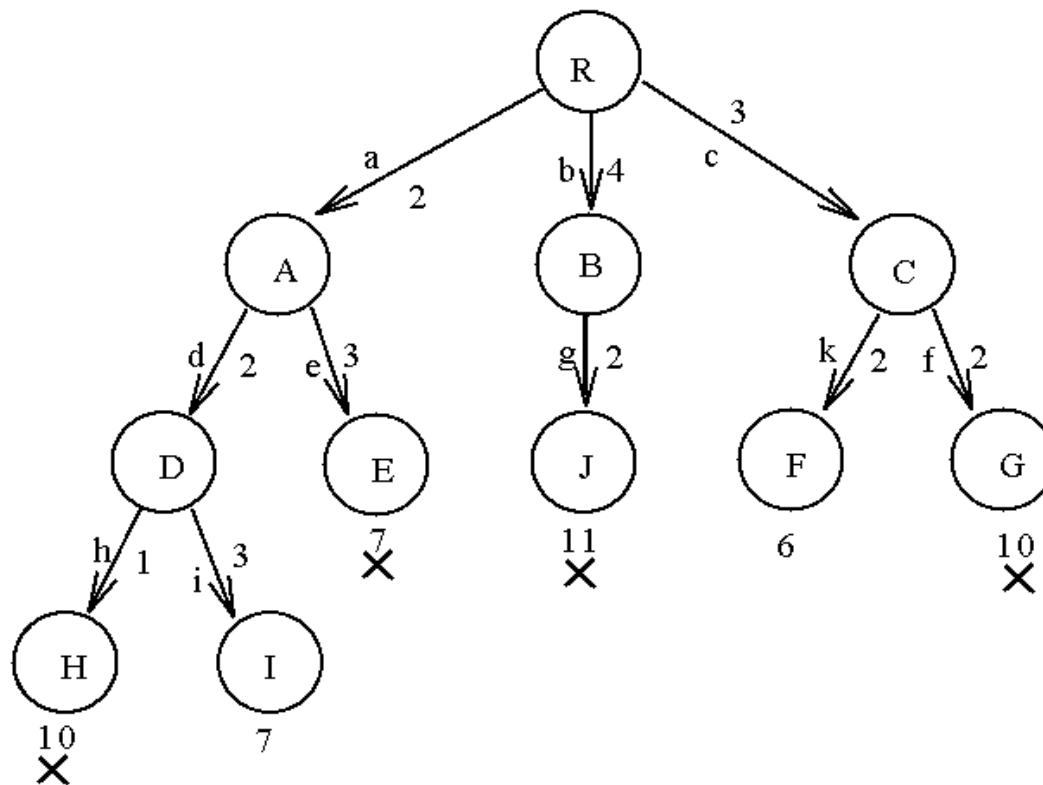


$$g(H)=2+2+1=5$$
$$g(I)=2+2+3=7$$

$$h(H)=\min\{5\}=5$$
$$h(I)=0$$

$$f(H)=5+5=10$$
$$f(I)=7+0=7^{23}$$

□ Step 5: Expand node B.

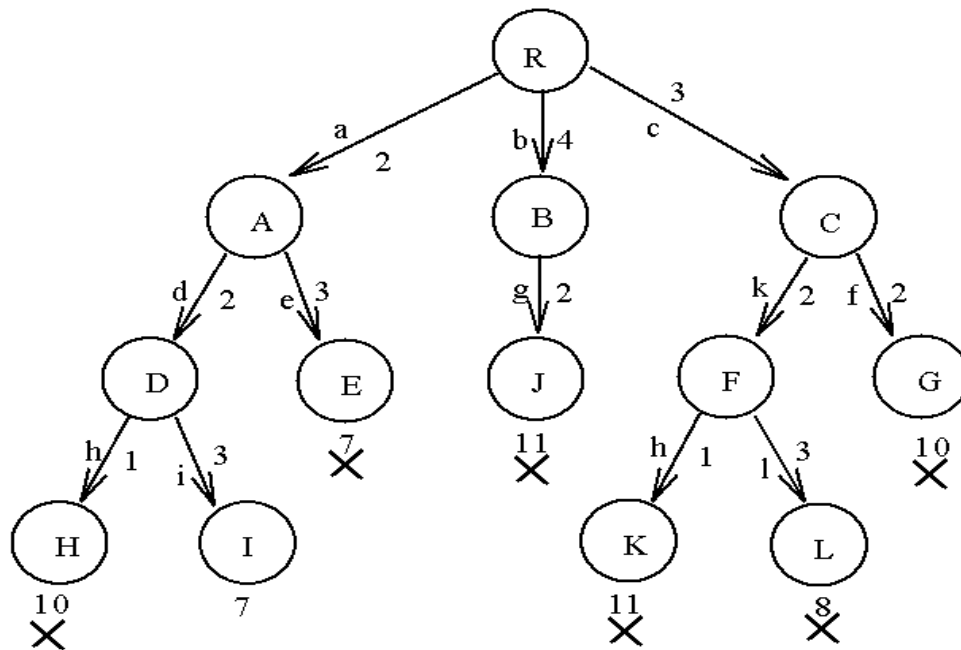


$$g(J)=4+2=6$$

$$h(J)=\min\{5\}=5$$

$$f(J)=6+5=11_{24}$$

□ Step 6: Expand node F.



$$f(n) \leq f^*(n)$$

$$g(K) = 3 + 2 + 1 = 6$$

$$h(K) = \min\{5\} = 5$$

$$f(K) = 6 + 5 = 11$$

$$g(L) = 3 + 2 + 3 = 8$$

$$h(L) = 0$$

$$f(L) = 8 + 0 = 8$$

Node I is a goal node. Thus, the final solution has been obtained.

Homework 1

- Solve the following sudoku puzzle.

			8		6			
				2			7	
		9	4			5		
4						3		6
3								
2	6				1	7		
9	3	5		1			4	
			5		8	2		
					3		1	